

Homework 1

MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control

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Problem 1. Model a mobile robot as a moving frame $\{r\}$ with $\hat{\mathbf{y}}_r$ pointing forward and $\hat{\mathbf{x}}_r$ pointing to the robot's right. All rotations are about $\hat{\mathbf{z}}$, positive counter-clockwise. The robot starts with $\{r\}$ coincident with the world frame $\{0\}$. It then executes, in order: translate forward 1 m; rotate 45° clockwise; translate forward 0.5 m; rotate 30° counter-clockwise; translate backward 2 m. Denote the resulting pose $\{\text{final}\}$. From $\{\text{final}\}$ the robot observes an object 1.5 m to its right; the object frame $\{\text{obj}\}$ has the same orientation as $\{\text{final}\}$.

- (a) Draw to scale, on a 0.5 m grid, the path taken by the robot and the location of the object.
- (b) Write ${}^0\mathbf{T}_{\text{final}}$ as an ordered product of $\text{Trans}(\cdot)$ and $\text{Rot}_z(\cdot)$ operators.
- (c) Calculate the homogeneous transformation matrix that would move the world origin to the robot's final pose, i.e., ${}^0\mathbf{T}_{\text{final}}$.
- (d) Calculate the homogeneous transformation matrix that would move the robot's final pose to the object, i.e., ${}^{\text{final}}\mathbf{T}_{\text{obj}}$.
- (e) Calculate the homogeneous transformation that would move the object to the world origin (assume the object is aligned with the robot), i.e., ${}^{\text{obj}}\mathbf{T}_0$.

Problem 2. Figure 1 shows a three-link manipulator. The first joint is at the origin O_0 and rotates about ${}^0\hat{\mathbf{z}}$ with an angle θ_0 . The first link has a length of 1 m. The second link is prismatic and extends and contracts with length a_1 , it is attached at a 45° rotation clockwise about the ${}^1\hat{\mathbf{z}}$ direction. The third joint is again rotary and rotates relative to the second link by angle θ_2 . The third link has a length of 1 m.

- (a) Draw the manipulator when $\theta_0 = \pi/4 \text{ rad}$; $a_1 = 0.5 \text{ m}$; $\theta_2 = \pi/4 \text{ rad}$, make sure you draw in and label the intermediate coordinate system directions.
- (b) Let θ_0 be the first joint's angle. What is the symbolic homogeneous transformation from the world frame to the distal (far) end of link 1, i.e., ${}^0\mathbf{T}_1$?

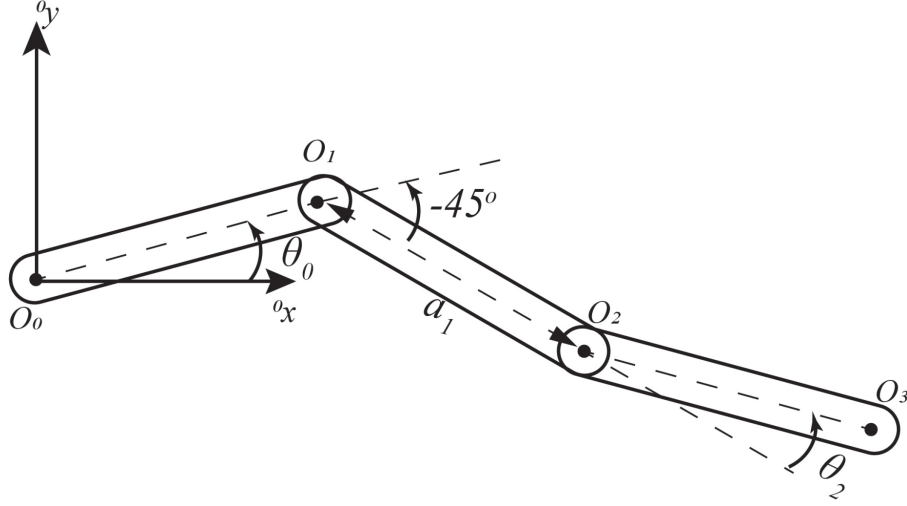


Figure 1: Three link RPR manipulator.

- (c) Let a_1 be the second joint's length. What is the symbolic homogeneous transformation from the distal end of link 1 to the distal end of link 2, i.e., ${}^1\mathbf{T}_2$?
- (d) Let θ_2 be the third joint's angle. What is the symbolic homogeneous transformation from the distal end of link 2 to the distal end of link 3, i.e., ${}^2\mathbf{T}_3$?
- (e) What is the symbolic homogeneous transformation from the world origin to the distal end of link 3, i.e., ${}^0\mathbf{T}_3$?

Problem 3. Figure 2 shows the path of a mobile robot. The displacements are given by:

$${}^0\mathbf{d}_{01} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, {}^1\mathbf{d}_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, {}^2\mathbf{d}_{23} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, {}^3\mathbf{d}_{34} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, {}^4\mathbf{d}_{45} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

- (a) Calculate: ${}^0\mathbf{T}_1$
- (b) Calculate: ${}^1\mathbf{T}_2$
- (c) Calculate: ${}^2\mathbf{T}_3$
- (d) Calculate: ${}^3\mathbf{T}_4$
- (e) Calculate: ${}^4\mathbf{T}_5$
- (f) Calculate: ${}^0\mathbf{T}_5$
- (g) Calculate: ${}^0\mathbf{d}_{05}$
- (h) Calculate: ${}^0\mathbf{R}_5$
- (i) Using homogeneous transformations,

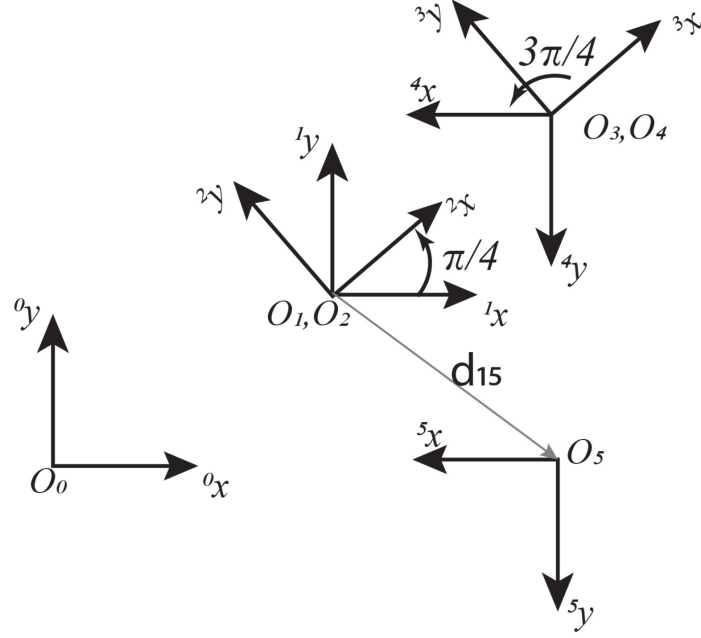


Figure 2: Path of mobile robot.

- (I) What is the equation in terms of homogeneous transformations to find the displacement vector from origin 1 to origin 5 (d_{15}) expressed in the final (Origin 5) frame?
- (II) What is this numerical displacement?

Problem 4. Determine if the following matrices are rotations and explain why or why not:

(a)
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(c)
$$\begin{bmatrix} -0.254 & 0.087 & -0.963 \\ 0.967 & 0.0169 & -0.253 \\ -0.00572 & -0.996 & -0.0884 \end{bmatrix}$$

(d)
$$\begin{bmatrix} 0.546 & 0.0719 & -1.47 \\ 0.814 & -0.696 & -0.0502 \\ -1.23 & -0.813 & 0.474 \end{bmatrix}$$

Problem 5. Complete the below rotation matrices. How do you know it is correct?

$$(a) \begin{bmatrix} 0 & 0 & ? \\ 0 & 1 & ? \\ 1 & 0 & ? \end{bmatrix}$$

$$(b) \begin{bmatrix} -1 & ? & ? \\ ? & 1 & ? \\ ? & ? & ? \end{bmatrix}$$

$$(c) \begin{bmatrix} 0.3698 & ? & 0.8665 \\ -0.9276 & ? & 0.324 \\ -0.05236 & ? & 0.3797 \end{bmatrix}$$

$$(d) \begin{bmatrix} ? & 0.1166 & ? \\ -0.2039 & ? & 0.2153 \\ ? & ? & -0.3672 \end{bmatrix}$$